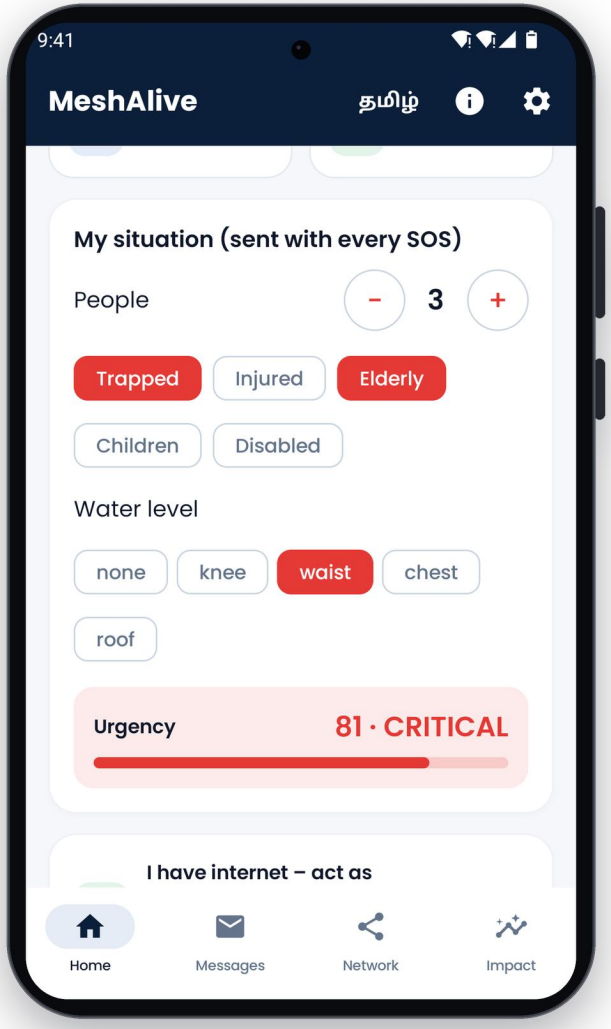
SEVAFIRST INNOVATION CHALLENGE 2026

MeshAlive

An Offline Disaster-Response Network

Carrying SOS messages to the district control room, and help back to people, when mobile towers fail.

Submission details	
Theme	05 · Surakshit Bharat
Track	Track A · Product or Technology
Stage	Working prototype
Website and demo	meshalive.benzo.co.in
Date	November 2026



CONTENTS

Agenda

01 Executive summary

02 The problem and who it affects

03 Existing approaches and their gaps

04 Proposed solution and key features

05 Prototype: mobile app and district control room

06 Technical architecture, routing and security

07 Evidence and results

08 Impact, cost and sustainability

09 Implementation roadmap and risks

10 Adoption, partners and support required

OVERVIEW

Executive summary

During cyclones and floods, Indian districts lose mobile coverage for hours or days, exactly when people most need help. MeshAlive keeps emergency communication working without towers, internet or new hardware.

- A one-tap SOS carries GPS location and a situation report, scored 0–100 for urgency.
- Nearby phones relay it over Bluetooth and Wi-Fi using battery-aware routing.
- Rescue boats and teams carry messages out of areas with no route.
- The first phone with signal uploads everything to a live district dashboard.
- Dispatch and evacuation messages travel back through the same mesh, digitally signed.

Status at a glance

Prototype

Working Android app and dashboard

Languages

Tamil and English

Testing

49 automated tests passing

Simulation

100% delivery, 200 phones

Hardware cost

₹0 for citizens and rescuers

Ask

District pilot partnership

THE PROBLEM

When the towers go down, rescue goes blind

Every cyclone and flood season, coastal and low-lying districts lose mobile coverage. Towers lose power, backhaul links are cut and the remaining network is overloaded. This happened in the 2015 Chennai floods and again during Cyclone Michaung in 2023.

People cannot call for help

Trapped families, the elderly, patients and people with disabilities cannot call, text or share their location.

Control rooms work blind

There is no view of who needs help, where they are, how many people are with them, or which case is most urgent.

Rescue capacity is misdirected

Boats and teams search by sight and word of mouth, so they reach the easiest-to-see rather than the most critical.

Problem statement

When cyclones and floods take down mobile towers and power, people in danger cannot call for help, and district control rooms cannot see who needs rescue, where they are, or how urgent it is.

THE PROBLEM

Who is affected, and where

People

- Families in flood- and cyclone-prone areas, especially the elderly, children, people with disabilities and patients
- Fishing communities and low-lying coastal settlements
- Rescue teams: NDRF, SDRF, fire and rescue services, boat operators and ambulance crews
- District Emergency Operations Centres and Disaster Management Authorities

Places

- Coastal districts of Tamil Nadu, such as Chennai, Cuddalore, Nagapattinam and Thoothukudi
- India’s cyclone-exposed east and west coasts
- River-flood regions: the Brahmaputra and Ganga basins, and Kerala
- Cities where flooding cuts power to mobile towers

Why it matters: in the first hours of a disaster, rescue capacity is scarce. Location and urgency data decide who is reached first.

CONTEXT

Existing approaches and their gaps

Approach	What it does	Gap during a disaster
Mobile calls and SMS	Standard emergency communication	Fails when towers lose power or backhaul
Offline chat apps (e.g. Bridgefy, Briar)	Phone-to-phone messaging	Do not reach rescuers, no prioritisation, no reply channel
LoRa radios (e.g. Meshtastic, goTenna)	Long-range text over radio	Extra hardware of ₹2,500+ per person; low adoption
Satellite phones	Works without local networks	Expensive; held by few officials, not by citizens
Broadcast alerts (cell broadcast, sirens)	One-way warnings	Cannot receive SOS or confirm who is safe

The gap: no low-cost, two-way channel connects people in danger to the district control room when the network is down.

PROPOSED SOLUTION

A complete rescue loop on phones people already own

MeshAlive is an Android app and a browser-based district dashboard. Together they form a two-way emergency channel that works without towers, internet or extra hardware.

1

Ask for help

One-tap SOS with GPS and a situation report, scored 0–100

2

Phones relay

Bluetooth and Wi-Fi mesh, battery-aware, most urgent first

3

Rescuers carry

Boats and teams carry SOS out of areas with no route

4

Control room sees

First phone with signal uploads to a live dashboard

5

Help returns

Dispatch and evacuation alerts travel back, signed

- No towers, no internet and no SIM needed for the mesh
- No extra hardware: uses citizens’ and rescuers’ existing phones
- Works in Tamil and English, with large, simple controls

PROPOSED SOLUTION

Key features

For people in danger

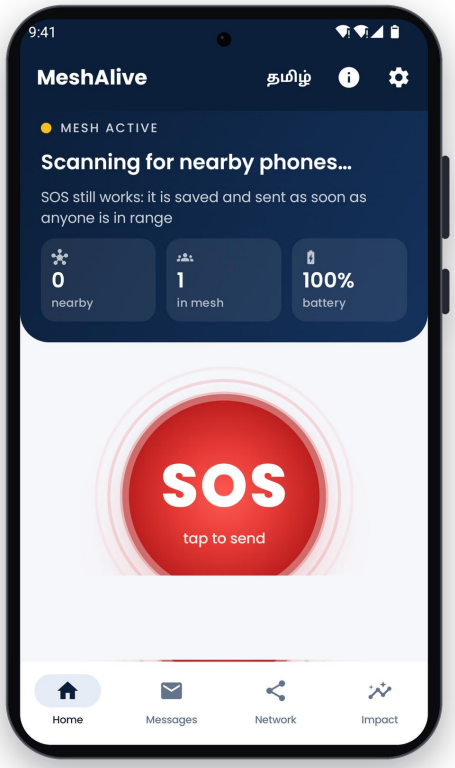
- 30-second guided setup: language, household and role
- One-tap SOS with a 3-second countdown and Cancel to prevent false alarms
- Situation report: people, trapped, injured, elderly, children, disabled and water level, turned into an urgency score
- “I’m safe” check-ins, so rescuers can skip safe households
- “Rescue team on the way” confirmation, even with no signal
- Full SOS flow in Tamil and English

For rescuers and officials

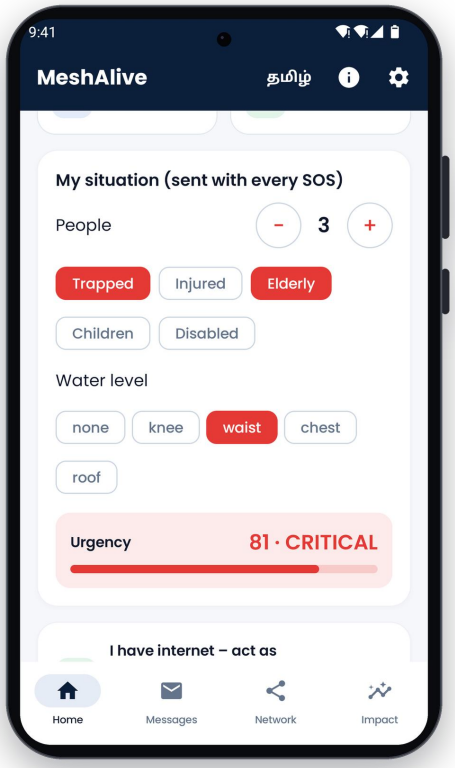
- Rescue-carrier mode for boats, NDRF/SDRF teams and ambulances
- Any phone that regains signal becomes a gateway automatically
- Live district dashboard: map, urgency, people count and route
- Dispatch team, mark rescued and evacuation broadcast, digitally signed
- District coverage planner: installs needed per ward
- Field-trial evidence recorder with exportable reports

PROTOTYPE

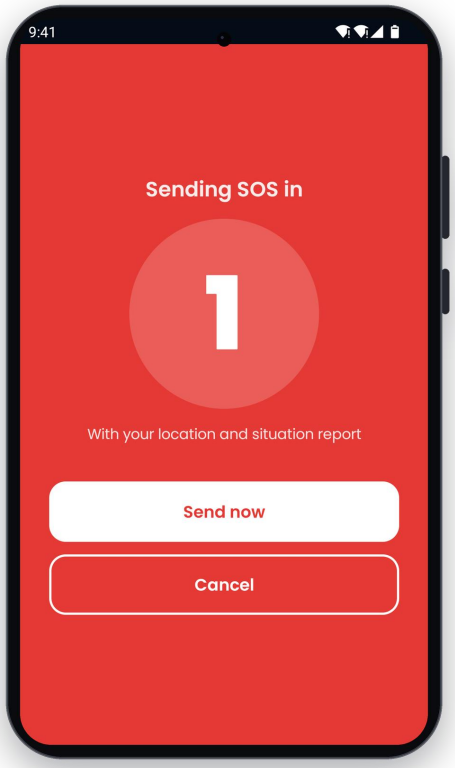
Mobile app: working Android prototype



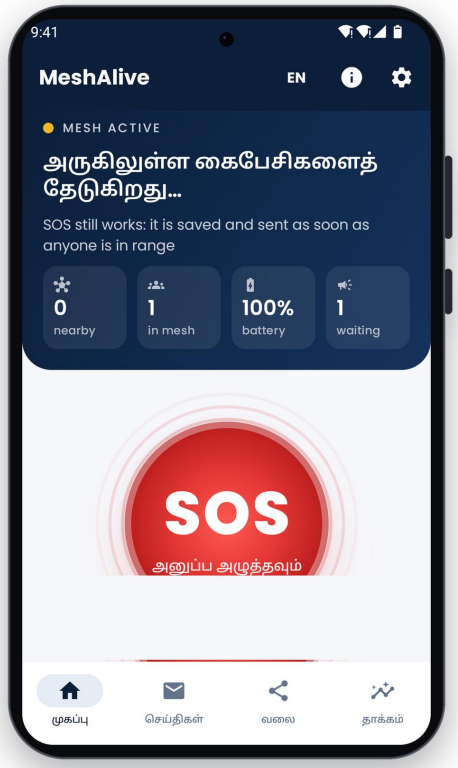
Home: mesh status, one-tap SOS



Situation report: urgency 81



3-second countdown

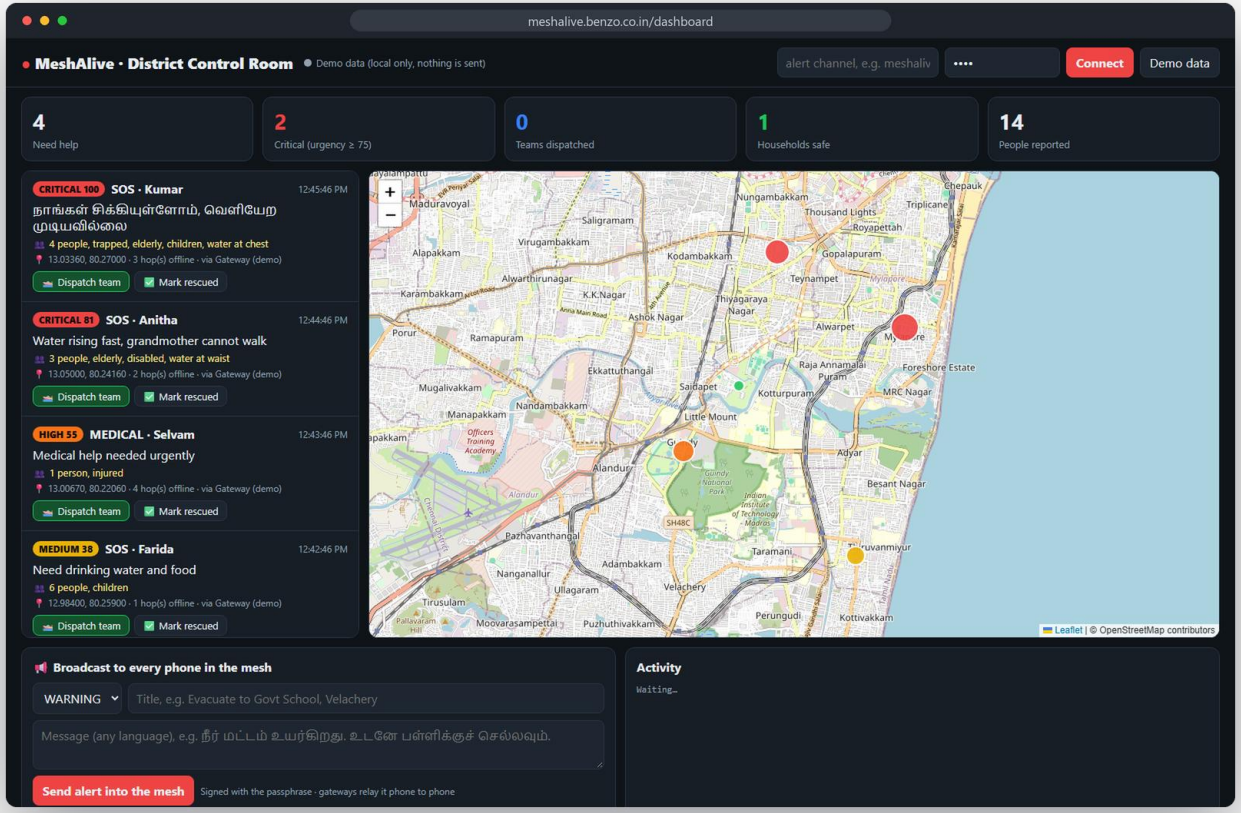


Complete flow in Tamil

Built with Kotlin and Jetpack Compose. All screens are captured from the running app.

PROTOTYPE

District control room dashboard



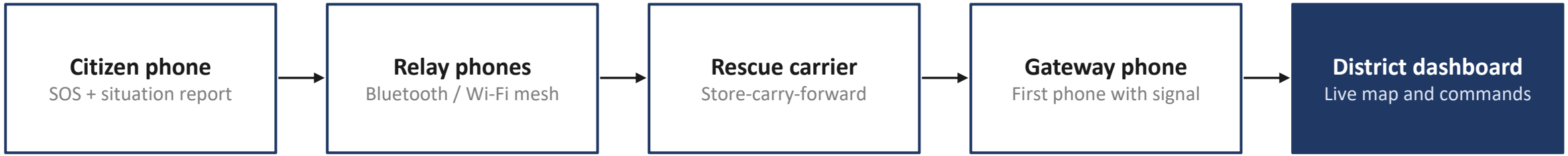
Operator actions

- **Live map sorted by urgency**
- Critical cases first, with people count and route
- **Dispatch team**
- Reply travels back; the family's phone confirms receipt
- **Mark rescued**
- Closes the case and informs the household
- **Evacuation broadcast**
- Any language, to every phone in the mesh

Runs in any browser on an existing control-room PC. Live demo: meshalive.benzo.co.in/dashboard

TECHNOLOGY

System architecture



Return path: signed dispatch and evacuation commands travel back through the gateway and the mesh to the citizen's phone

Layer	Implementation
Mobile app	Kotlin, Jetpack Compose, Room database, Android 8.0 or later
Peer-to-peer link	Google Nearby Connections (Bluetooth and Wi-Fi), no internet required
Cloud relay	Lightweight publish/subscribe relay (prototype: ntfy), self-hostable
Dashboard	Browser app (HTML/JavaScript, Leaflet maps), no installation

TECHNOLOGY

Routing and reliability

Mechanism	How it works	Benefit
Network map	Each phone shares battery, neighbours and internet status every few seconds; link quality measured by round-trip time	Every phone knows the live mesh
Battery-aware routing	Dijkstra with edge cost = link quality + battery penalty + hop cost; a virtual super-node joins all gateways	Avoids phones about to die; one run, O((V+E) log V)
Urgency priority queue	Messages sent in order of triage score	Most critical SOS go first
Store-and-forward	Messages kept and retried when no route exists	Nothing lost while out of range
Duplicate filter and TTL	Seen-message cache and hop limit	No loops or broadcast storms
Delivery receipts	Acknowledgement returned with hop count	Sender knows the SOS arrived
Rescue carriers	Custody transfer to boat and team phones	Reaches areas with no route at all

Security and privacy

Controls in the prototype

- AES-GCM encryption of message content on the phone
- Control-room commands signed with HMAC-SHA256
- Phones reject unsigned, altered or expired commands (6-hour limit)
- No account or sign-up; no personal data beyond what the SOS needs
- Location shared only inside an SOS sent by the user

Threats addressed

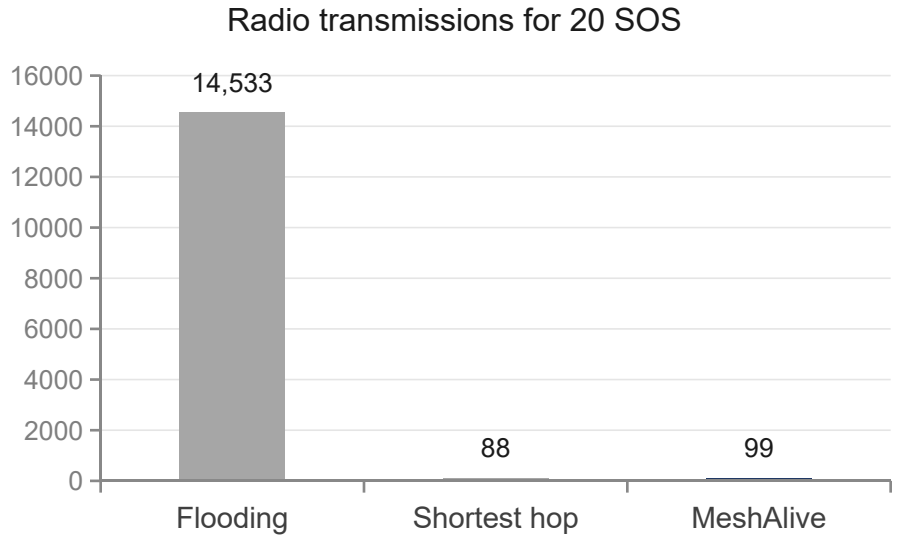
Threat	Mitigation
Fake evacuation order	Signature check on every command
Replay of old orders	Timestamp and expiry window
Message flooding	Duplicate filter and hop limit
Accidental SOS	3-second countdown with Cancel

EVIDENCE

Results: 200-phone simulation

Measured with the app's own routing engine: 200 phones, 20 SOS messages, the same network for every method.

Measure	Flooding	Shortest hop	MeshAlive
SOS delivered	85%	100%	**100%**
Radio transmissions	14,533	88	**99**
Average hops	5.47	4.40	**4.95**
Battery left, weakest phone	0.0%	3.2%	**6.3%**



- Every SOS delivered, with 147 times fewer transmissions than flooding
- Twice the battery left on the weakest phone compared with shortest-hop routing, so the mesh stays alive longer
- 49 automated tests cover routing, forwarding, triage, command signatures, the planner and evidence reports

INNOVATION

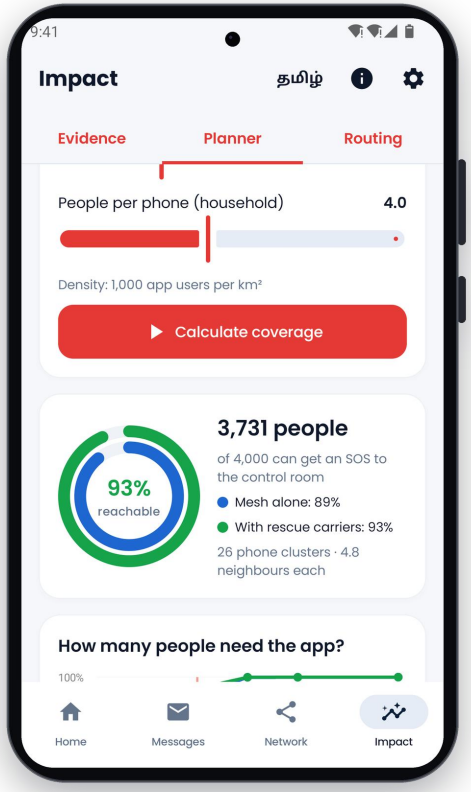
What is new

Capability	Chat apps	LoRa radios	MeshAlive
Reaches the district control room	No	No	**Yes, live dashboard**
Reply back to the person	No	No	**Dispatch and evacuation alerts**
Prioritises urgent cases	No	No	**Urgency score 0–100**
No phone in range	Message waits	Message waits	**Rescue carriers carry it out**
Battery management	Flooding	Flooding	**Battery-aware routing**
Protection against fake alerts	Not addressed	Not addressed	**Signed commands**
Extra hardware	None	₹2,500+ per person	**None**
Indian languages	English	English	**Tamil and English**

Beneficiaries and expected impact

Who benefits

- People trapped or stranded: their SOS reaches rescuers without towers, and they know help is coming
- Vulnerable groups: prioritised automatically through the situation report
- Rescue teams: exact location, people count and urgency, instead of searching blind
- Control rooms: a live, prioritised picture, and a channel to reach people without signal
- Administration: “I’m safe” check-ins reduce wasted searches



Coverage model

~3,700

of 4,000 people can reach the control room in a 1 km² ward with 1,000 app users (about 4 people per phone).

Planner model; to be validated in the pilot.

IMPACT

Cost and sustainability

Stakeholder	Hardware cost	Notes
Citizens	₹0	Free app on their own phone; no SIM, data or account needed for the mesh
Rescue carriers	₹0	Uses phones that crews already carry
District control room	₹0	Runs in any browser on an existing PC
Running cost	Low	Small self-hosted relay server and app maintenance
Rollout	Programme cost	Pre-monsoon awareness and installation drives through the DDMA

Sustainability

- Funded through district disaster-preparedness budgets; no per-user licence
- Open, auditable code that state IT departments can host and maintain

IMPLEMENTATION

Implementation roadmap

Phase 1 · Field pilot 3 months

- One flood-prone ward with the DDMA
- NDRF/SDRF boats as rescue carriers
- Dashboard in the district EOC
- Measure delivery rate and SOS-to-dispatch time

Phase 2 · District rollout Pre-monsoon

- Install drives via schools, colleges, panchayats and SHGs
- Planner sets the install target per ward
- Training for control-room operators

Phase 3 · State and national Scale-up

- Distribution through the SDMA
- Dashboard feeds into state EOCs
- Alignment with the Common Alerting Protocol
- More Indian languages; iOS version

Success measures: SOS delivery rate, time from SOS to dispatch, households confirmed safe, and installs per ward.

IMPLEMENTATION

Risks and mitigations

Risk	Impact	Mitigation
Too few phones with the app in an area	Messages cannot hop far	Planner sets install targets; rescue carriers bridge gaps; pre-monsoon install drives
Battery drain on relaying phones	Mesh shrinks over time	Battery-aware routing; low-battery phones avoided
Fake or malicious alerts	Panic, misdirected rescue	Signed commands; expiry window; duplicate filtering
Accidental or prank SOS	Wasted rescue effort	Countdown with Cancel; urgency triage; “I’m safe” check-ins
Dependence on a public relay	Availability and data control	Self-hosted relay on government infrastructure for deployment
Android-only today	Excludes iPhone users	iOS version planned for Phase 3

WAY FORWARD

Adoption, partners and support required

Who could adopt it

- District Disaster Management Authorities and EOCs (primary users)
- State Disaster Management Authorities, such as TNSDMA
- NDMA, for guidelines and Common Alerting Protocol integration
- NDRF, SDRF and fire and rescue services, as rescue carriers
- Urban local bodies, panchayats, NGOs, NSS and NCC units

Support we are seeking

- A pilot partnership with a district administration and its DDMA
- Collaboration with NDRF/SDRF and boat operators to test carrier mode
- Mentorship in disaster-management operations and public-sector deployment
- Technical review: security, Common Alerting Protocol, self-hosting
- Pilot funding for a relay server, test devices and training drives

CONCLUSION

Thank you

MeshAlive turns the phones people already own into a rescue network, so that when the mobile network fails, help does not.

Resource	Link
Website and live demo	meshalive.benzo.co.in
Control-room dashboard	meshalive.benzo.co.in/dashboard
Android app (APK)	meshalive.benzo.co.in/download/MeshAlive.apk
Demo video	meshalive.benzo.co.in/#demo
Source code	github.com/yogeshwardev/MESHALIVE-

